

Hritvik Arora

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EDUCATION

Indian Institute of Science Education and Research (IISER) Kolkata <i>BS-MS Dual Degree Programme</i>	2024 – Present <i>CGPA: 7.91/10</i>
City Montessori School, Lucknow <i>ISC (Class XII)</i>	2021 – 2023 <i>94%</i>
City Montessori School, Lucknow <i>ICSE (Class X)</i>	2019 – 2021 <i>96.4%</i>

RESEARCH INTERESTS

- Computational Geoscience • Computational Seismology • Geophysics • Scientific Computing
- Numerical Modelling • Seismic Imaging • Scientific Image Processing • Digital Petrography
- Quantitative Microstructural Analysis • Structural Geology

RESEARCH EXPERIENCE

Summer Research Intern <i>Indian Institute of Technology Roorkee</i>	May 2026 – June 2026 <i>Supervisor: Prof. Sudheer Kumar Tiwari</i>
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- Assisted in computational analysis of petrographic datasets for structural geology applications.
- Performed MATLAB-based scientific programming and data analysis for geological applications.
- Prepared technical documentation and research reports related to ongoing research activities.

PROJECTS

GrainSegmenter – MATLAB based Software for Petrographic Image Analysis

- Developed a MATLAB-based software tool for semi-automated grain boundary segmentation of petrographic thin section images.
- Designed an interactive graphical interface supporting manual boundary refinement through draw/erase tools, adjustable segmentation tolerance, undo/redo operations, zoom controls, and project persistence.
- Reduced manual effort in petrographic image analysis by integrating automated segmentation with efficient user guided boundary refinement.

MYflow Software Validation and Enhancement (MATLAB)

- Performed algorithmic validation of the MYflow composite-mode workflow to investigate grain count inconsistencies in petrographic phase-map analysis.
- Improved the grain identification pipeline by redesigning boundary detection from the original RGB image and enhancing phase assignment, connected-component analysis, and overall workflow robustness.
- Developed multiple reliability improvements including input validation, duplicate-phase protection, zero-label filtering, histogram edge-case handling, and enhanced error management.

- Authored a technical validation report documenting the investigation, code modifications, and experimental evaluation. The work was acknowledged and appreciated by Dr. Leonardo Casini, developer of MYflow.

TECHNICAL SKILLS

Programming: MATLAB, Python, Java, LaTeX

Scientific Computing: Image Processing, Digital Image Analysis, Numerical Modelling, Data Analysis

Software: MATLAB, QGIS, Git, GitHub, Adobe Illustrator, Microsoft Excel, SnapGene

POSITIONS OF RESPONSIBILITY

Treasurer — Cricket Club, IISER Kolkata

- Managed club finances and coordinated tournaments.

Editor — PREP4IISER

- Formatted previous-year examination papers and solutions using LaTeX.
- Maintained publication-quality formatting across educational material.

Sports and Extra-Curricular

- An active institute level Cricketer.
- Represented IISER Kolkata as a Wicket Keeper Batsman in Inter Institute Sports Meet 2025.

RELEVANT COURSEWORK

Structural Geology, Geophysics, Petrology, Mineralogy, Numerical Methods, Advanced Programming in Python, Remote Sensing, Earth System Science.